ALFA RIZI

Software Engineer | Backend / Full Stack

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SUMMARY

Entry-level Software Engineer with hands-on experience building backend systems, full-stack applications, distributed services, and AI-powered applications through professional, academic, and independent projects. Experienced with Java, Spring Boot, Quarkus, React, Next.js, Python, PostgreSQL, Docker, and CI/CD.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, Python, SQL, C
Backend: Spring Boot, Quarkus, Spring Cloud Gateway, FastAPI, Express.js
Frontend: React, Next.js, Tailwind CSS, Vite, Web Speech API
Database & Infrastructure: PostgreSQL, Redis, ChromaDB, Docker, Docker Compose
Tools: Git, GitHub Actions, Linux, OpenAPI/Swagger

PROFESSIONAL EXPERIENCE

Lead Intern — IT & Business Operations | PT Resika Cemerlang (Pasim Group), Bandung | 2025·Sep 2026

- Led a 5-person intern team supporting daily IT operations, hardware diagnostics, and workstation SLA compliance.
- Managed OS security patching, software configuration, and antivirus definition updates across corporate laptops.
- Validated technical specifications across thousands of medical device asset records, maintaining database integrity and audit readiness.

LEADERSHIP & ORGANIZATIONS

Academic Staff — HIMAMI

- Coordinated academic development activities, peer-to-peer coding support, and technical learning initiatives.

SELECTED ENGINEERING PROJECTS

ReLoop — Enterprise Circular-Commerce Platform & Escrow Architecture
Java 17, Quarkus 3.15 LTS, Next.js 14, PostgreSQL 16, Redis 7 | 2026

- Architected a modular monolith across 25 bounded contexts, enforcing module boundaries with ArchUnit and asynchronous processing through a transactional outbox worker.
- Engineered a double-entry escrow ledger and transaction-safe checkout leases using PostgreSQL pessimistic locking (SELECT ... FOR UPDATE) to prevent concurrent inventory conflicts.
- Integrated Next.js with TanStack Query and Redis catalog caching for responsive product browsing and inventory-aware checkout.

Demo: reloop.biz.id | GitHub: github.com/a1fariz/ReLoop
ApexGrid — F1 Race Ticket Reservation & Microservices Platform
Java 17, Spring Boot 3.2, Spring Cloud Gateway, PostgreSQL, React 18, Docker | 2026

- Architected a 3-service distributed system (Gateway, User, Race) with Spring Cloud Gateway and stateless JWT-based RBAC security.
- Synchronized ticket inventory with PostgreSQL PL/pgSQL triggers and an automated 30-minute reservation-expiry worker.
- Built a React admin analytics panel with OpenAPI/Swagger documentation and containerized the platform with Docker Compose.

GitHub: github.com/a1fariz/ApexGrid
StudyMate AI — Asynchronous RAG Document Knowledge Assistant
Python 3.12, FastAPI, ChromaDB, Gemini 2.0 Flash, Docker | 2026

- Built an asynchronous RAG pipeline that parses textbook PDFs with PyMuPDF, performs vector search with ChromaDB, and returns exact page citations.
- Integrated Gemini 2.0 Flash for summaries, quizzes, and 3D flashcards, with a 60 requests/minute token-bucket rate limiter.

GitHub: github.com/a1fariz/StudyMateAI
Vanguard IELTS — Academic Preparation & SRS Platform
React 19, TypeScript, Vite, Tailwind CSS v4, Web Speech API | 2026

- Developed an IELTS preparation suite covering Listening, Reading, Writing, and Speaking across CEFR A1–C1 levels.
- Implemented a SuperMemo-inspired spaced-repetition vocabulary engine and Shadowing Studio using native Web Speech synthesis.

Demo: vanguard-ielts.vercel.app | GitHub: github.com/a1fariz/Vanguard-IELTS

EDUCATION

Universitas Nasional PASIM — Associate Degree in Informatics Management (D3) | GPA: 3.78/4.00
Academic Status: Coursework completed; currently preparing for final thesis defense (sidang).
PUB Intensive Technical Scholarship — Logic & C, Data Structures, Relational Databases, Advanced Java Development, and React Development | 2024–Present

CERTIFICATIONS & TECHNICAL TRAINING

- Intensive React Front-End Development Training — PUB Programming Training Center, PASIM National University (Sep 2026)
- Fundamental Back-End Programming Training (Java Programming Language) — PUB Programming Training Center, PASIM National University (Jan 2026)
- Web Programming Training (HTML, CSS, JavaScript) — PUB Programming Training Center, PASIM National University (Aug 2025)
- Database Training — PUB Programming Training Center, PASIM National University (May 2025)
- Data Structure Training — PUB Programming Training Center, PASIM National University (May 2025)
- Git and GitHub Training — PUB Programming Training Center, PASIM National University (Jan 2026)